2016 A Level Paper 2 Suggested Solutions

1	**Suggested Solutions**
	Let V be volume of water (in m^3). Let h be depth of water (in m). Always define new variables. Given $\dfrac{dV}{dt} = 0.1$ Find $\dfrac{dh}{dt}$ when $V = 3$ $\dfrac{dh}{dt} = \dfrac{dh}{dV} \times \dfrac{dV}{dt}$ $\quad \left(\text{to find } \dfrac{dh}{dV}, \text{ need } V \text{ in terms of } h \right)$ Chain rule $V = \dfrac{1}{3}\pi r^2 h$ From diagram, $\dfrac{r}{h} = \tan\alpha$ Toa Cah Soh $r = \dfrac{h}{2}$ Sub $r = \dfrac{h}{2}$ into V: Eliminate r before differentiating with respect to h. $V = \dfrac{1}{12}\pi h^3$ $\dfrac{dV}{dh} = \dfrac{1}{4}\pi h^2$ $\dfrac{dh}{dt} = \dfrac{dh}{dV} \times \dfrac{dV}{dt}$ $\quad = \dfrac{4}{\pi h^2} \times 0.1$ When $V = 3$, $3 = \dfrac{1}{12}\pi h^3$ $h = \left(\dfrac{36}{\pi}\right)^{\frac{1}{3}}$ Rate of increase of depth of water $= 0.025050 \approx 0.0251$ (3 s.f.)

2	Suggested Solutions

(a)(i)

$$\int x^2\cos nx\ \mathrm{d}x = x^2\frac{\sin nx}{n} - \int 2x\left(\frac{\sin nx}{n}\right)\mathrm{d}x$$

Integration by parts:
$u = x^2 \Rightarrow \dfrac{\mathrm{d}u}{\mathrm{d}x} = 2x$

$$= \frac{1}{n}x^2\sin nx - \frac{2}{n}\int x\sin nx\ \mathrm{d}x$$

Integration by parts:
$u = x \Rightarrow \dfrac{\mathrm{d}u}{\mathrm{d}x} = 1$

$$= \frac{1}{n}x^2\sin nx - \frac{2}{n}\left(x\left(\frac{-\cos nx}{n}\right) - \int\frac{-\cos nx}{n}\ \mathrm{d}x\right)$$

$$= \frac{1}{n}x^2\sin nx - \frac{2}{n}\left(-\frac{1}{n}x\cos nx + \frac{1}{n}\int\cos nx\ \mathrm{d}x\right)$$

$$= \frac{1}{n}x^2\sin nx - \frac{2}{n}\left(-\frac{1}{n}x\cos nx + \frac{1}{n}\left(\frac{\sin nx}{n}\right)\right) + c,\quad c\in\mathbb{R}$$

$$= \frac{1}{n}\left(x^2\sin nx + \frac{2}{n}x\cos nx - \frac{2}{n^2}\sin nx\right) + c$$

(a)(ii)

$$\int_\pi^{2\pi} x^2\cos nx\ \mathrm{d}x = \frac{1}{n}\left[x^2\sin nx + \frac{2}{n}x\cos nx - \frac{2}{n^2}\sin nx\right]_\pi^{2\pi}$$

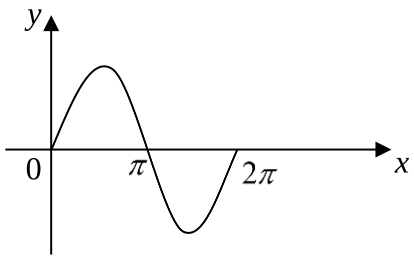
$$= \frac{1}{n}\left(0 + \frac{2}{n}(2\pi) - 0 - \left(0 + \frac{2}{n}\pi\cos n\pi - 0\right)\right)$$

$$= \frac{1}{n}\left(\frac{4\pi}{n} - \frac{2\pi}{n}\cos n\pi\right)$$

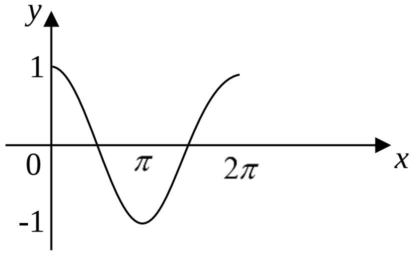
$$= \frac{2\pi}{n^2}(2 - \cos n\pi)$$

$$= \begin{cases}\dfrac{6\pi}{n^2} & \text{if } n \text{ is odd}\\[2mm] \dfrac{2\pi}{n^2} & \text{if } n \text{ is even}\end{cases}$$

Therefore, $a = \begin{cases}6 & \text{if } n \text{ is odd}\\ 2 & \text{if } n \text{ is even}\end{cases}$.



For any $n\in\mathbb{Z}$, $\sin nx = 0$



For any $n\in\mathbb{Z}$, $\cos 2nx = 0$

For any $n\in\mathbb{Z}$, $\cos nx = \begin{cases}1 & \text{if } n \text{ is even}\\ -1 & \text{if } n \text{ is odd}\end{cases}$

(b)

$$\text{Required volume} = \pi\int_0^2 y^2 \, \mathrm{d}x$$

$$= \pi\int_0^2 \frac{x^3}{\left(9-x^2\right)^2} \, \mathrm{d}x$$

$$= \pi\int_0^2 \left(-\frac{1}{2}\right)\frac{x^2}{\left(9-x^2\right)^2}(-2x) \, \mathrm{d}x$$

$$= \pi\int_9^5 \left(-\frac{1}{2}\right)\frac{9-u}{u^2} \, \mathrm{d}u$$

$$= \frac{\pi}{2}\int_5^9 \frac{9}{u^2}-\frac{1}{u} \, \mathrm{d}u$$

$$= \frac{\pi}{2}\left[-\frac{9}{u}-\ln|u|\right]_5^9$$

$$= \frac{\pi}{2}\left(-1-\ln 9+\frac{9}{5}+\ln 5\right)$$

$$= \frac{\pi}{2}\left(\frac{4}{5}+\ln\frac{5}{9}\right) \text{ units}^3$$

$u = 9 - x^2 \Rightarrow x^2 = 9-u$

$\frac{\mathrm{d}u}{\mathrm{d}x} = -2x$

When $x = 2,\ u = 5$.

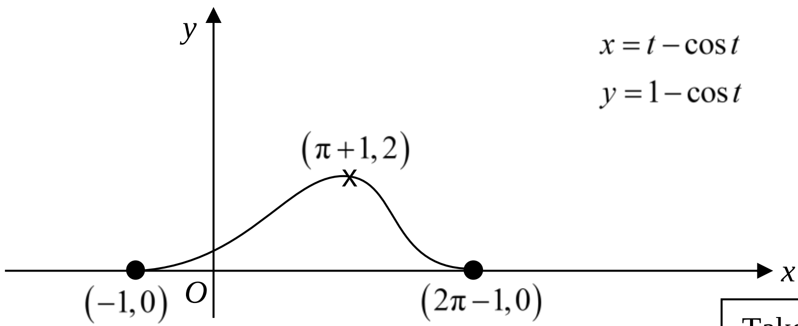
When $x = 0,\ u = 9$.

Sub all of the following in 1 step:
1) Expression
2) dx
3) Limits

Swap limits.
$\int_a^b \mathrm{f}(x)\,\mathrm{d}x = -\int_b^a \mathrm{f}(x)\,\mathrm{d}x$

3	Suggested Solutions

(i)



$x = t - \cos t$

$y = 1 - \cos t$

Take note:
1) Exact coordinates
2) Label end-points, ie. $0 \le t \le 2\pi$

$\frac{\mathrm{d}x}{\mathrm{d}t} = 1 + \sin t$ $\qquad$ $\frac{\mathrm{d}y}{\mathrm{d}t} = \sin t$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\sin t}{1+\sin t}$

when $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$,

$\sin t = 0$

$t = 0, \pi, 2\pi$

when $t = \pi, x = \pi + 1, y = 2$

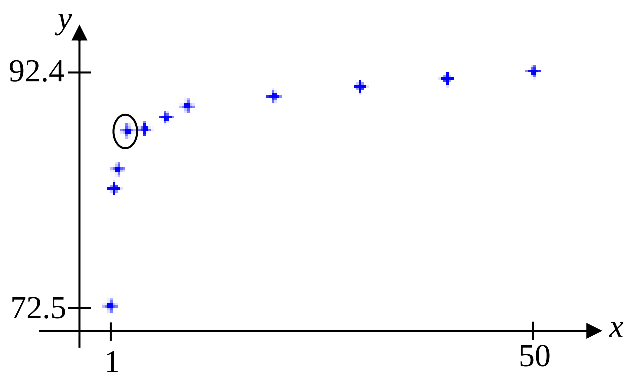
(ii)	**Method 1: Using $\int \mathbf{f}'(x)\mathbf{f}(x)\mathrm{d}x$** Area under $D = \int_{-1}^{a-\cos a} y \ \mathrm{d}x$ $= \int_0^a (1-\cos t)(1+\sin t) \ \mathrm{d}t$ $= \int_0^a 1-\cos t+\sin t-\sin t\cos t \ \mathrm{d}t$ $= \left[t-\sin t-\cos t-\frac{1}{2}\sin^2 t\right]_0^a$ $= a-\sin a-\cos a-\frac{1}{2}\sin^2 a-(0-0-1-0)$ $= a-\sin a-\cos a-\frac{1}{2}\sin^2 a+1$ **Method 2: Using $\sin 2x = 2\sin x\cos x$** Area under $D = \int_{-1}^{a-\cos a} y \ \mathrm{d}x$ $= \int_0^a (1-\cos t)(1+\sin t) \ \mathrm{d}t$ $= \int_0^a 1-\cos t+\sin t-\sin t\cos t \ \mathrm{d}t$ Double Angle Formula $= \int_0^a 1-\cos t+\sin t-\frac{1}{2}\sin 2t \ \mathrm{d}t$ $= \left[t-\sin t-\cos t+\frac{1}{4}\cos 2t\right]_0^a$ $= a-\sin a-\cos a+\frac{1}{4}\cos 2a+\frac{3}{4}$	$x = t-\cos t$ $\frac{\mathrm{d}x}{\mathrm{d}t} = 1+\sin t$ When $t=a,\ x=a-\cos a$ When $t=0,\ x=-1$ $\int \mathrm{f}'(t)\mathrm{f}(t)\mathrm{d}t = \frac{\mathrm{f}(t)}{2}+C$ $\int \sin t\cos t \,\mathrm{d}t = \frac{\sin^2 t}{2}+C$ $\mathrm{f}(t)=\sin t$ $\mathrm{f}'(t)=\cos t$
(iii)	$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y}{\mathrm{d}t}\times\frac{\mathrm{d}t}{\mathrm{d}x}$ $= \frac{\sin t}{1+\sin t}$ When $t=\frac{\pi}{2},\ x=\frac{\pi}{2}, y=1, \frac{\mathrm{d}y}{\mathrm{d}x}=\frac{1}{2}$. Gradient of normal $= \frac{-1}{\frac{1}{2}} = -2$ Equation of normal: $y-1=-2\left(x-\frac{\pi}{2}\right)$ At $x=0,\ y=1+\pi$. At $y=0,\ x=\frac{\pi+1}{2}$. Area of $\Delta DEF = \frac{1}{2}(1+\pi)\left(\frac{\pi+1}{2}\right) = \left(\frac{\pi+1}{2}\right)^2$	

5	Suggested Solutions
(i)	$$\begin{array}{l} \qquad\qquad\quad \text{R} \;\; * \quad \text{B} \;\; * \quad \text{Y} \;\; * \\ \text{P}(\text{win game}) = \frac{4}{7}\times\frac{1}{6}+\frac{2}{7}\times\frac{2}{6}+\frac{1}{7}\times\frac{3}{6} \\ \qquad\qquad\quad = \frac{11}{42} \end{array}$$
(ii)	$$\text{P}(\mathbf{B}\mid\text{win game}) = \frac{\text{P}(\mathbf{B}\text{ and win game})}{\text{P}(\text{win game})}$$ $$\begin{array}{l} \quad \text{B} \;\; * \\ = \dfrac{\frac{2}{7}\times\frac{2}{6}}{\frac{11}{42}} = \frac{4}{11} \end{array}$$
(iii)	$$\begin{array}{l} \qquad\qquad\qquad\qquad\qquad\qquad \text{R} \;\; * \quad \text{B} \;\; * \quad \text{Y} \;\; * \\ \text{P}(\text{win consecutive games}) = \left(\frac{4}{7}\times\frac{1}{6}\right)\left(\frac{2}{7}\times\frac{2}{6}\right)\left(\frac{1}{7}\times\frac{3}{6}\right)3! \\ \qquad\qquad\qquad\qquad\qquad = \frac{4}{1029} \end{array}$$ Permute 3 different colours

6	Suggested Solutions
(iii)	Let X be the age of a randomly chosen employee (in years). Let μ denotes the population mean age of employees (in years). H_0: $\mu = 37$ H_1: $\mu < 37$ Under H_0, since n = 80 is large, by Central Limit Theorem, $\bar{X} \sim N\left(37, \frac{140}{80}\right)$ approximately. Test statistic: $Z = \dfrac{\bar{X}-37}{\sqrt{140/80}} \sim N(0,1)$ Level of significance: 5% Reject H_0 if z-value < -1.6449 Since the Managing Director's belief ($\mu < 37$) should be accepted at 5% significance level, H_0 should be rejected. Under H_0, z-value < -1.6449 $\dfrac{\bar{x}-37}{\sqrt{140/80}} < -1.6449$ $\bar{x} < 34.824$ $\therefore \; \{\bar{x} \in \mathbb{R} : 0 < \bar{x} < 34.8\}$

(iv)	Let X be the age of a randomly chosen employee (in years). Let μ denotes the population mean age of employees (in years). H_0: $\mu = 37$ H_1: $\mu < 37$ Under H_0, since n = 80 is large, by Central Limit Theorem, $\bar{X} \sim N\left(37, \frac{140}{80}\right)$ approximately. Test statistic: $Z = \frac{\bar{X} - 37}{\sqrt{140/80}} \sim N(0,1)$ Level of significance: $\alpha\%$ Reject H_0 if p-value < $\frac{\alpha}{100}$ Under H_0, using G.C, p-value = 0.086809 (5 s.f.) Since the Managing Director's belief should not be accepted at $\alpha\%$ significance level, H_0 should not be rejected. p-value > $\frac{\alpha}{100}$ $0.086809 > \frac{\alpha}{100}$ $\alpha < 8.6809$ $\therefore\ \{\alpha \in \mathbb{R}^+ : 0 \le \alpha < 8.68\}$

7	**Suggested Solutions**
(i)	Choose 3 out of the 4 women Permute the 3 chosen women Number of ways = ${}^4C_3 \times 3! = 24$
(ii)	Number of ways = **No restriction – All men – All women** = ${}^{10}C_3 \times 3! - {}^6C_3 \times 3! - 24 = 576$
(iii)	**For numerator, group the 3 in one group so we have 8 units to arrange at a round table. Remember to arrange within the group as well.** Permute 8 units at a round table Permute within the grouped unit Required Probability = $\dfrac{\frac{8!}{8} \times 3!}{\frac{10!}{10}} = \dfrac{1}{12}$
(iv)	**For numerator, arrange the remaining 7 persons. Next, choose 3 slots out of the 7 available slots and arrange the 3 of them into the 3 chosen slots.** Required Probability = $\dfrac{\frac{7!}{7} \times {}^7C_3 \times 3!}{\frac{10!}{10}} = \dfrac{5}{12}$

8	**Suggested Solutions**
(i)	
(ii)	As x increases, y increases at a decreasing rate. Therefore $y = ax + b$ is not a suitable model.
(iii)	From the scatter diagram: Since y increases as x increases, c is negative. Since efficiency is non-negative, d is positive.
(iv)	From GC, $r = -0.97955 \approx -0.980 \quad (3 \text{ s.f.})$ $c = -17.484 \approx -17.5 \quad (3 \text{ s.f.})$ $d = 91.750 \approx 91.8 \quad (3 \text{ s.f.})$ Regression line: y on $\frac{1}{x}$ $y = 91.750 - \frac{17.484}{x}$ $y = 91.8 - \frac{17.5}{x} (3 \text{ s.f.})$
(v)	When $x = 3$ $y = 91.750 - \frac{17.484}{3} = 85.922 \approx 85.9$ The estimated value = $85.9 \quad (3 \text{ s.f.})$ Since $r = -0.980$ is close to -1, which indicates a strong negative linear correlation between y and $\frac{1}{x}$ and $x = 3$ is within the data range of x, the estimate is reliable.

9	Suggested Solutions

(a)

$X \sim \mathrm{N}\left(15, a^2\right)$

$\mathrm{P}\left(10 < X < 20\right) = 0.5$

Method 1: (Graphical Method)
From GC,
$a = 7.4130 \approx 7.41 \ \left(3 \text{ s.f.}\right)$

Method 2: (Standardisation)

$$\mathrm{P}\left(\frac{10-15}{a} < X < \frac{20-15}{a}\right) = 0.5$$

$$\frac{20-15}{a} = 0.6744897$$

$$a = 7.4130 \approx 7.41 \text{ (3 s.f.)}$$

(b)

$Y \sim \mathrm{B}\left(4, p\right)$

Distribution of X	$\mathrm{P}(X = x)$
Binomial $\mathrm{B}(n,p)$	$\binom{n}{x} p^x (1-p)^{n-x}$

$\mathrm{P}\left(Y = 1\right) + \mathrm{P}\left(Y = 2\right) = 0.5$

$$\binom{4}{1} p\left(1-p\right)^3 + \binom{4}{2} p^2\left(1-p\right)^2 = 0.5$$

$$4p\left(1-3p+3p^2-p^3\right) + 6p^2\left(1-2p+p^2\right) = 0.5$$

$$8p - 24p^2 + 24p^3 - 8p^4 + 12p^2 - 24p^3 + 12p^4 = 1$$

$$4p^4 - 12p^2 + 8p = 1 \quad \text{(Shown)}$$

Using G.C.,

$p = -2.01$ or $p = 0.166$ or $p = 0.599$ or $p = 1.25$

$\left(\text{rej} \because p > 0\right)$ $\left(\text{rej} \because p < 1\right)$

$\therefore p = 0.166$ or $p = 0.599$